ME352 QUBE Motor Lab 2
Motor Speed PI Control

Zeshui Song

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1 Overview

In this lab we will apply feedback control to the QUBE DC motor system with the goal of acceptable transient response and tracking. Proportional control (P) offers the advantage of fast transient response and reducing steady state error. While higher gains yield better performance, they come at the expense of increased sensitivity to noise. We will learn that integral control (I) offers the advantage of zero steady-state error for a step input to this system. (This is not generally true for all systems). Unfortunately, however, as the gain of I is increased to improve rise time, overshoot and instability will also increase. We will see that combining P and I (PI control) is better than either P or I alone by simultaneously offering the advantages of quick transient response, zero steady-state error and decreased sensitivity to noise.

2 Goals

Our hands-on goals for this lab are to:

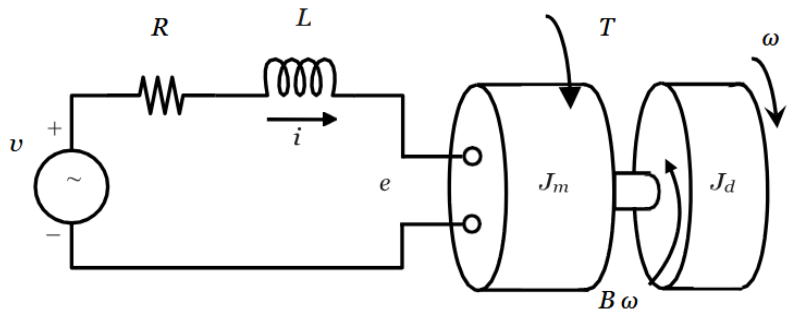
- Learn how to work as a designer. Use a model of the DC motor system to design a suitable controller which is implemented in an experiment.
- Design a feedback controller for reference tracking using P, I, and PI control approaches.
- Compare measured responses to those simulated for the modeled system.

3 Pre-lab Questions

Come prepared to your scheduled lab session with the pre-lab questions answered. Turn in the pre-lab questions with the lab assignment.

Transfer Function Derivation

Schematic



Where:

- $e = k_b\omega$ is the back EMF voltage (law of the generator)
- $T = k_m i$ is the torque generated by the motor (law of the motor)

Equations of Motion
Assuming a rigid connection between the motor and the load disc, the equations of motion for the system are:

KVL:

$$L\frac{di}{dt} + Ri + e = V \quad \Longrightarrow \quad L\frac{di}{dt} + Ri + k_b\omega = V$$

Sum of torques (Newton):

$$J\frac{d\omega}{dt} + B\omega = T \quad \Longrightarrow \quad J\frac{d\omega}{dt} + B\omega = k_m i$$

Where $J = J_m + J_d$ is the total inertia of the system.

Laplace Transform
With zero initial conditions since we are solving for the transfer function:

$$L(sI(s)) + RI(s) + k_b\Omega(s) = V(s) \quad \Longrightarrow \quad I(s) = \frac{V(s) - k_b\Omega(s)}{Ls + R}$$

$$J(s\Omega(s)) + B\Omega(s) = k_m I(s)$$

Plug in for $I(s)$:

$$J(s\Omega(s)) + B\Omega(s) = k_m\frac{V(s) - k_b\Omega(s)}{Ls + R}$$

Rewrite:

$$Js\Omega(s)(Ls + R) + B\Omega(s)(Ls + R) = k_m V(s) - k_m k_b\Omega(s)$$

$$(JLs^2 + JRs)\Omega(s) + (BLs + BR)\Omega(s) + k_m k_b\Omega(s) = k_m V(s)$$

$$\boxed{\Longrightarrow \; G(s) = \frac{\Omega(s)}{V(s)} = \frac{k_m}{JLs^2 + (JR + BL)s + (BR + k_m k_b)} \quad \text{(transfer fxn)}}$$

Parameter Identification

Recall from lab 1 that the first-order transfer function of the motor plant is given by:

$$P(s) = \frac{K}{\tau s + 1}$$

1. Write down the calculated values for K and τ you determined in lab 1, including units, using the QUBE-Servo manufacturer values of

- Motor resistance, $R = 8.4\ \Omega$,
- Motor inertia (includes the hub attachment inertia), $J_m = 4.65 \times 10^{-6}\ \mathrm{kg\,m^2}$,
- Motor torque constant, $K_m = 0.042\ \mathrm{N\,m/A}$,
- Motor back EMF constant, $K_b = 0.042\ \mathrm{V/(rad/s)}$.

To calculate the load disc inertia, J_d, assume the disc has a mass of 0.053 kg and radius of 0.0248 m. Assume the motor damping/friction and the armature inductance are negligible.

Assumptions for a first order system

- Neglect the motor inductance, $L = 0$.
- Neglect the motor damping/friction, $B = 0$.

$$G(s) = \frac{k_m}{JRs + k_m k_b}$$

Compare to the form of a first-order transfer function:

$$G(s) = \frac{K}{\tau s + 1} = \frac{k_m}{JRs + k_m k_b} = \frac{\frac{k_m}{k_m k_b}}{\frac{JR}{k_m k_b} s + 1}$$

Thus:

- DC gain:

$$K = \frac{k_m}{k_m k_b} = \frac{1}{k_b}$$

- Time constant:

$$\tau = \frac{JR}{k_m k_b}$$

Solve for J_d

$$J_d = \frac{1}{2} m_d r_d^2 = \frac{1}{2}[0.053\ \mathrm{kg}][0.0248\ \mathrm{m}]^2 = 1.629 \times 10^{-5}\ \mathrm{kg\,m^2}$$

$$\implies J = J_m + J_d = [4.65 \times 10^{-6}\ \mathrm{kg\,m^2}] + [1.629 \times 10^{-5}\ \mathrm{kg\,m^2}] = 2.094 \times 10^{-5}\ \mathrm{kg\,m^2}$$

Plug in and solve for parameters

- DC gain:

$$K = \frac{1}{k_b} = \frac{1}{[0.042\ \text{V/(rad/s)}]} = \boxed{23.809\ \text{(rad/s)/V}}$$

- Time constant:

$$\tau = \frac{JR}{k_m k_b} = \frac{[2.094 \times 10^{-5}\ \text{kg m}^2][8.4\ \Omega]}{[0.042\ \text{N m/A}][0.042\ \text{V/(rad/s)}]} = \boxed{0.0997\ \text{s}}$$

Proportional (P), Integral (I) and Proportional-Integral (PI) Control

A common block diagram for reference tracking is shown in Fig. 1. Recall that a proportional-integral controller has transfer function:

$$C(s) = k_p + \frac{k_i}{s}$$

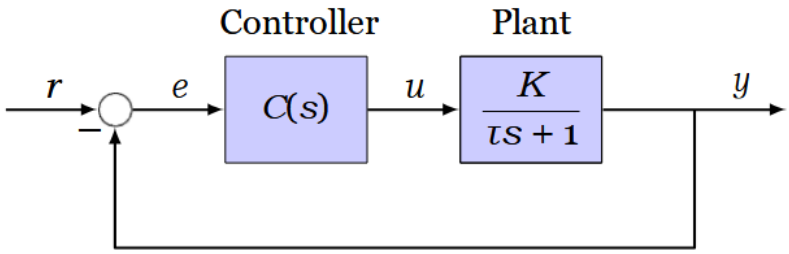


Figure 1: Block diagram for error-based feedback control (reference tracking).

In the following, we consider proportional, integral and proportional-integral control for the motor-tachometer system. For all three types of control our goal is to design a controller such that:

(i) the rise time is less than 0.2 seconds,

(ii) the steady-state error is less than 1%,

(iii) the overshoot is less than 20%,

(iv) the noise in the input signal (applied motor voltage) is small. The spread of the amplitude with respect to the mean signal should be less than 0.25 volts.

You, the control designer, need to find out if you can meet all prescribed specifications.

For this lab, use the approximation that rise time from 10–90% of steady-state is

$$t_r \approx \frac{1.8}{\omega_n}.$$

Note this is only a rough approximation based on the behavior of second-order systems with no zeros. For first-order systems, the transient behavior can be better characterized by the time constant, which is related to the pole location.

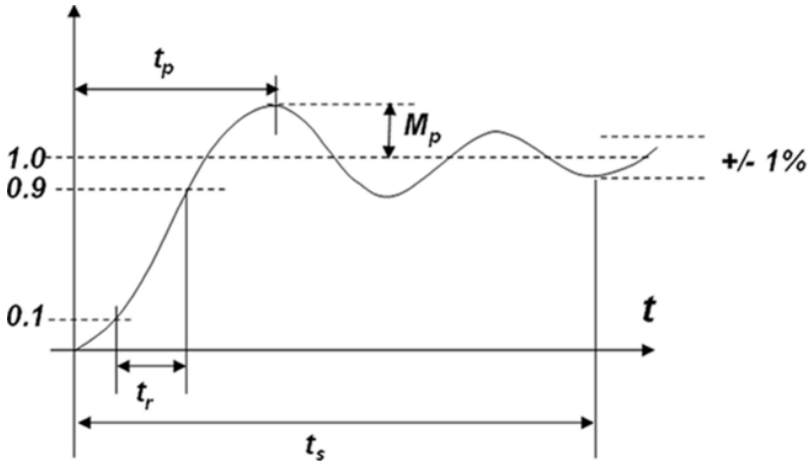
You may recall or will learn that overshoot is

$$M_p(\zeta) = \frac{y_{\text{peak}} - y_{\text{ss}}}{y_{\text{ss}}} = e^{-\pi\zeta/\sqrt{1-\zeta^2}},$$

which approximately yields $\zeta > 0.5$ for $M_p < 20\%$.

Note the form:

$$\frac{s}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \quad \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$



Proportional Control

First, we consider proportional control, i.e.

$$C(s) = k_p.$$

2. Use the final value theorem and reference-to-error transfer function, or determine the error from system type and error constants, to write the steady-state error to a unit step input in terms of the system parameters K and/or τ and the proportional control constant k_p.

Error Transfer Function:

$$\frac{E(s)}{R(s)} = \frac{1}{1 + C(s)P(s)} = \frac{1}{1 + \frac{k_p K}{\tau s + 1}}$$

$$\frac{E(s)}{R(s)} = \frac{\tau s + 1}{\tau s + 1 + k_p K}$$

Final value theorem with a unit step input:

$$R(s) = \frac{1}{s}$$

$$\lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s) = \lim_{s\to 0}\left(s\cdot\frac{\tau s+1}{\tau s+1+k_pK}\cdot\frac{1}{s}\right) = \boxed{\frac{1}{1+k_pK}}$$

3. Design P-controllers such that the steady state error is respectively less than 30%, 10%, and 1%. Calculate the respective ranges of k_p values to meet each steady-state error specification.

The steady-state error is given by

$$e_{ss} = \frac{1}{1+k_pK}$$

Since this was derived for a unit step input ($r_{ss} = 1$), the percentage of steady-state error is

$$\%e_{ss} = \frac{y_{ss}-r_{ss}}{r_{ss}}\cdot 100 = \frac{e_{ss}}{r_{ss}}\cdot 100 = e_{ss}\cdot 100$$

- For $e_{ss} < 30\%$:

$$\frac{1}{1+k_pK} < 0.3 \implies 1+k_pK > \frac{1}{0.3} \implies \boxed{k_p > \frac{7}{3K} = 0.098 \text{ V/(rad/s)}}$$

- For $e_{ss} < 10\%$:

$$\frac{1}{1+k_pK} < 0.1 \implies 1+k_pK > \frac{1}{0.1} \implies \boxed{k_p > \frac{9}{K} = 0.378 \text{ V/(rad/s)}}$$

- For $e_{ss} < 1\%$:

$$\frac{1}{1+k_pK} < 0.01 \implies 1+k_pK > \frac{1}{0.01} \implies \boxed{k_p > \frac{99}{K} = 4.158 \text{ V/(rad/s)}}$$

4. Find the closed-loop pole and closed-loop time constant for the k_p value corresponding to 1% steady-state error.

Closed Loop Transfer Function

$$\frac{Y(s)}{R(s)} = \frac{C(s)P(s)}{1+C(s)P(s)} = \frac{\frac{k_pK}{\tau s+1}}{1+\frac{k_pK}{\tau s+1}} = \frac{k_pK}{\tau s+1+k_pK}$$

Pole

$$\tau s+1+k_pK = 0 \implies \boxed{s = -\frac{1+k_pK}{\tau}}$$

For k_p corresponding to 1% steady-state error, $k_p = 99/K$:

$$s = -\frac{1+99}{\tau} = \boxed{-\frac{100}{\tau}}$$

Closed Loop Time Constant
Comparing the closed loop transfer function with the form of a first-order system:

$$\frac{Y(s)}{R(s)} = \frac{k_p K}{\tau s + 1 + k_p K} = \frac{K_{cl}}{\tau_{cl} s + 1}$$

$$\implies \tau_{cl} = \frac{\tau}{1 + k_p K} = \boxed{\frac{\tau}{100}}$$

Recall that the plant/open loop time constant is $\tau = 0.0997$ s from earlier.

Proportional-Integral Control

In the proportional-integral controller case,

$$C(s) = k_p + \frac{k_i}{s},$$

we can combine the favorable features of both P and I control, namely quick response, zero steady state error, and low sensitivity to noise.

5. Find the closed-loop transfer function for P-I control. Which property of the closed-loop system is influenced by k_p? Similarly, for k_i?

Closed Loop Transfer Function

$$C(s)P(s) = \left(k_p + \frac{k_i}{s}\right)\frac{K}{\tau s + 1} = \frac{K(k_p s + k_i)}{s(\tau s + 1)}$$

$$\frac{Y(s)}{R(s)} = \frac{C(s)P(s)}{1 + C(s)P(s)} = \frac{\frac{K(k_p s + k_i)}{s(\tau s + 1)}}{1 + \frac{K(k_p s + k_i)}{s(\tau s + 1)}} = \frac{K(k_p s + k_i)}{s(\tau s + 1) + K(k_p s + k_i)}$$

$$\boxed{\frac{Y(s)}{R(s)} = \frac{K k_p s + K k_i}{\tau s^2 + (1 + K k_p)s + K k_i}}$$

Compare the demoninator with the standard form of a second-order system:

$$\tau s^2 + (1 + K k_p)s + K k_i = s^2 + 2\zeta\omega_n s + \omega_n^2$$

Thus,

$$\boxed{\omega_n = \sqrt{\frac{K k_i}{\tau}}}$$

$$\zeta = \frac{1 + K k_p}{\tau} \cdot \frac{1}{2\omega_n} = \frac{1 + K k_p}{2\tau} \cdot \sqrt{\frac{\tau}{K k_i}}$$

$$\boxed{\zeta = \frac{1 + K k_p}{2\sqrt{\tau K k_i}}}$$

So k_p primarily affects the damping ratio, ζ, and k_i primarily affects the natural frequency, ω_n. Both gains affect the stability of the system.

6. For your PI-controller design, find the range of k_p and k_i values that meet or exceed the rise time $t_r \leq 0.2$ s, overshoot $M_p \leq 20\%$, and steady-state error less than 1%.

Note, if you did the calculations correctly, you should see the k_p gain required to meet the specifications using P-control only is much higher than in the PI-control case. In fact, the theoretical calculations should reveal that you should be able to meet specifications (i)–(iii) with I-control only. During the lab we will see if this is possible in practice.

Rise Time
Using the approximation for a 10–90% rise time:

$$t_r \approx \frac{1.8}{\omega_n} \leq 0.2 \implies \omega_n \geq 9 \text{ rad/s}$$

$$\implies \frac{Kk_i}{\tau} \geq 81 \implies \boxed{k_i \geq \frac{81\tau}{K}}$$

Plugging in the values for τ and K:

$$\implies k_i \geq \frac{[81 \text{ (rad/s)}^2][0.0997 \text{ s}]}{[23.809 \text{ (rad/s)/V}]} = \boxed{0.339 \text{ V/rad}}$$

Overshoot
The lab says that an overshoot of less than 20% corresponds to a damping ratio of $\zeta > 0.5$. Thus:

$$\zeta = \frac{1 + Kk_p}{2\sqrt{\tau K k_i}} > 0.5$$

$$1 + Kk_p > \sqrt{\tau K k_i}$$

$$k_p > \frac{\sqrt{\tau K k_i} - 1}{K} = \frac{\sqrt{[0.0997 \text{ s}][23.809 \text{ (rad/s)/V}][0.339 \text{ V/rad}]} - 1}{[23.809 \text{ (rad/s)/V}]}$$

$$\boxed{k_p > -0.00432 \text{Vs/rad}}$$

Steady State Error
The error transfer function is

$$\frac{E(s)}{R(s)} = \frac{1}{1 + C(s)P(s)} = \frac{1}{1 + \frac{K(k_p s + k_i)}{s(\tau s + 1)}} = \frac{s(\tau s + 1)}{s(\tau s + 1) + K(k_p s + k_i)}$$

Final value theorem with a unit step input $R(s) = \frac{1}{s}$:

$$\lim_{t \to \infty} e(t) = \lim_{s \to 0} sE(s) = \lim_{s \to 0} \left(s \cdot \frac{s(\tau s + 1)}{s(\tau s + 1) + K(k_p s + k_i)} \cdot \frac{1}{s} \right) = \frac{0}{K k_i} = \boxed{0}$$

Thus, the steady-state error is zero for a PI-controller, and the k_p and k_i gains can be chosen to meet the rise time and overshoot specifications.

Using Only k_i
It is possible to meet the specifications using only k_i:

- Rise time: $k_i \geq 0.339$ V/rad
- Overshoot:

$$k_p = 0 \implies \frac{1}{2\sqrt{\tau K k_i}} > 0.5 \implies k_i < \frac{1}{\tau K}$$

$$k_i < \frac{1}{[0.0997 \text{ s}][23.809 \text{ (rad/s)/V}]} = \boxed{0.421 \text{ V/rad}}$$

- Steady-state error: $e_{ss} = 0$ for any $k_i > 0$

4 Lab Assignment: Tracking the Motor Speed

The theoretical motor model used for the lab is

$$P(s) = \frac{K}{\tau s + 1}.$$

For this lab, use the manufacturer parameters for K and τ when comparing the theoretical model with the actual motor response.

Proportional Control Implementation

Now let us implement proportional control with the k_p values calculated in the pre-lab assignment.

7. Implement the k_p values from (3), with

$$k_i = 0.$$

Discuss how the response changes as you increase k_p and whether you are able to meet the specifications for (i)–(iii).

For a reference input of 50 rad/s:

- $k_p = 0.098$ V/(rad/s) for 30% steady-state error:
  - Rise time (s): 93.516 ms
  - Steady-state error (%): 30.422%
- $k_p = 0.378$ V/(rad/s) for 10% steady-state error:
  - Rise time (s): 18.792 ms
  - Steady-state error (%): 10.222%

- $k_p = 4.158$ V/(rad/s) for 1% steady-state error:
  - Rise time (s): 22.255 ms
  - Overshoot (%): 17.28%
  - Steady-state error (%): Oscillatory but the mean steady-state error is about 1%

Generally, because the closed loop system is first-order for proportional control, there was zero overshoot for all the lower k_p values where there were no oscillations. Increasing the gain k_p decreased the rise time and steady-state error, however this is not the case for high k_p values. It actually amplified the noise and caused oscillations in the response.

8. Verify in an experiment whether you can meet all design specifications (i)–(iv).

The noise in the input signal, the applied motor voltage, should be small. The spread of the amplitude with respect to the mean signal should be less than 0.25 volts.

Do the controllers operate as you expect? Why or why not?

Response
For a proportional gain of $k_p = 4.158$ V/(rad/s), the value corresponding to 1% steady-state error, the rise time and overshoot specifications were met. However, the oscillations were not given enough time to stabilize, as was seen with values close to 4.158 V/(rad/s). Additionally, the mean signal spread of the input signal was 204.305 V, which is much larger than the desired 0.25 V. The noise in the input signal was amplified by the high k_p gain, which caused oscillations in the response. Particularly, we saw sharp spikes in the input voltage at the beginning and end of the signal.

This was unexpected because, theoretically, the closed-loop system is first-order and should not exhibit oscillations. However, the effect of noise in the system was more significant than anticipated.

Integral Control Implementation

Next, we consider integral control only, i.e.

$$C(s) = \frac{k_i}{s}.$$

9. Implement an integral controller with

$$k_p = 0$$

using the specifications

$$t_r \leq 0.2 \text{ s},$$

less than 1% steady-state error, and overshoot less than 20%.

The noise in the input signal, the applied motor voltage, is desired to be small. The spread of the amplitude with respect to the mean signal should be less than 0.25 volts.

Can you meet all design specifications (i)–(iv)? Why or why not? Do the controllers operate as you expect based on your theoretical model?

Response
For an integral gain of $k_i = 0.42$ V/rad, we found the following system time specifications:

- Rise time (s): 170.024 ms
- Overshoot (%): 29.87%
- Steady-state error (%): 0.26%
- Voltage spread (V): 2.23 V

Thus, the controller was able to meet the rise time and steady-state error specifications, but it did not meet the overshoot or input signal spread specifications. The large input signal spread could potentially be due to noise in the system, as with proportional control, although integral control was much better at limiting the input signal spread. The overshoot was higher than desired, which is consistent with the fact that the system becomes second-order with integral control and can exhibit an underdamped response.

The controller operated close to as expected, but differences between the actual system and the theoretical model were mainly due to noise and other effects in the physical system that were not accounted for in the theoretical model.

Proportional-Integral Control Implementation

Now, let us implement the proportional-integral controller.

10. Keeping integral control, start with a very low k_p gain and gradually increase k_p to see the change in response with a large k_p gain. Observe how increasing k_p affects the response.

Vary both gains to see if you can meet the specifications from (i)–(iv).

The noise in the input signal, the applied motor voltage, should be small. The spread of the amplitude with respect to the mean signal should be less than 0.25 volts.

How can k_p and k_i be traded off with each other to achieve the desired system behavior?

List your final k_p and k_i and whether you were able to meet all the specifications.

Response

$$k_p = 0.08 \text{ V/(rad/s)}$$

$$k_i = 0.42 \text{ V/rad}$$

We found the following system time specifications:

- Rise time (s): 208.04 ms
- Overshoot (%): $\approx 0\%$
- Steady-state error (%): 0.018%
- Voltage spread (V): 3.09 V

This controller was able to meet the overshoot and steady-state error specifications, getting very close to the rise time specification. However, the input signal spread was still larger than desired. The closed loop response behaved closer to a first order system with oscillations appearing with higher k_p gains. Potentially, it is due to the non-zero k_p gain that resulted in a larger input signal spread than with purely integral control.

Overall, the gains k_p and k_i can be traded off to achieve a similar system response. This can be seen mathematically since the damping ratio ζ is dependent on both k_p and k_i, while the natural frequency ω_n is dependent only on k_i. Thus, the closed-loop response became more heavily damped as k_p increased, as predicted by the theoretical model. However, higher values of k_p also produced greater high-frequency oscillations in the experimental response. This was likely due to the proportional gain amplifying noise and other effects that were not included in the theoretical model.

5 Effect of Pole Location on Response

11. Use the following MATLAB code to plot the root locus for the system with P-control. Sketch or turn in a MATLAB plot of the root locus plot.

The $\times$ should show the location of the open-loop pole, which should be

$$s = -\frac{1}{\tau}.$$

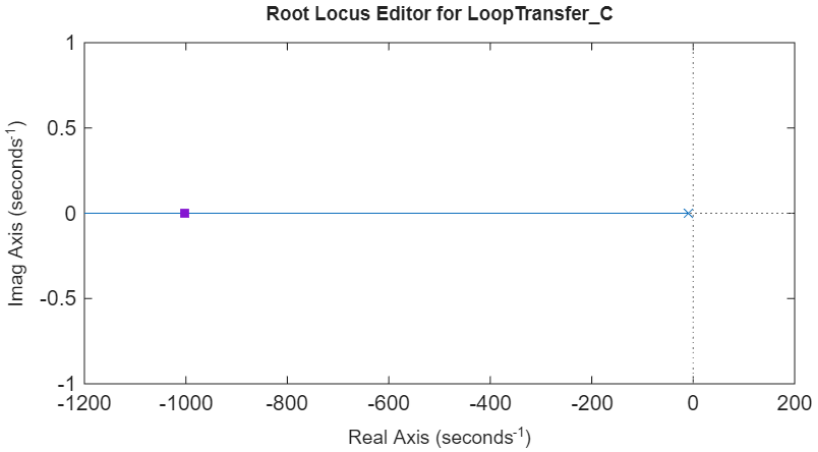
```
s = tf('s');
sys = K /(tau*s + 1);
rlocus(sys)
```

The root-locus design method uses the open-loop transfer function, in this case $C(s)P(s)$ since the system has unity feedback, and indicates the open-loop poles and zeros.

Either with hand calculations or using MATLAB, root locus branches can be plotted which show the locations of all possible closed-loop poles as you vary one parameter in the system, usually a controller gain, between 0 and ∞.

Try using the k_p value corresponding to 1% error from Question 4. The value of the closed-loop pole should agree with your calculations in Question 4.

Root Locus Plot:



This checks out since the open loop pole marked with an $\times$ is at $s = -\frac{1}{\tau} = -10.03$ rad/s, and the closed-loop pole marked with an $\circ$ is at $s = -\frac{1+k_pK}{\tau} = -10002.98$ s^{-1}, which agrees with the calculations in Question 4.

12. Add MATLAB code or modify the system to plot the root locus for the system with I-control.

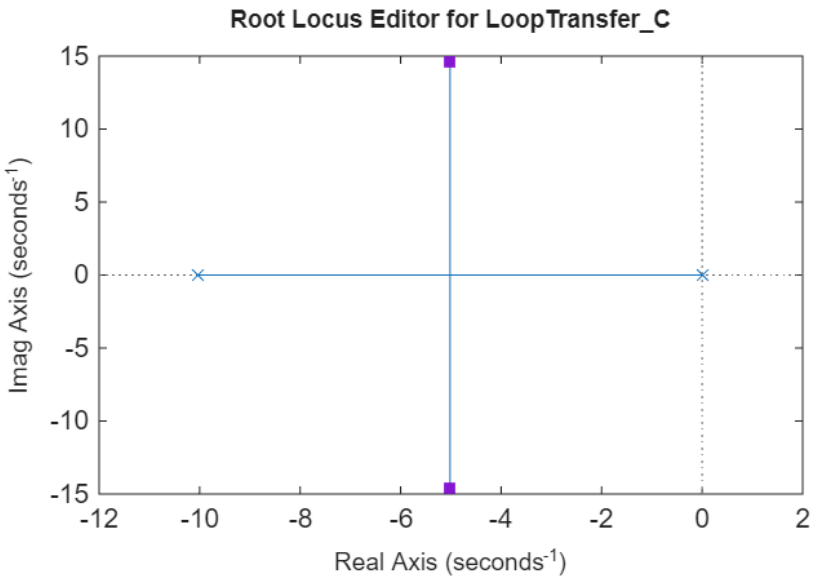
Remember to use the open-loop transfer function

```
sys = K/(s*(tau*s + 1))
```

Sketch or turn in a MATLAB plot of the root locus for the system with I-control.

Do not include the k_i gain, as this is the parameter being varied in the root locus plot.

Root Locus Plot:



13. What happens to the root loci branches and corresponding closed-loop poles as you increase k_i and how does this affect the closed-loop time response characteristics?

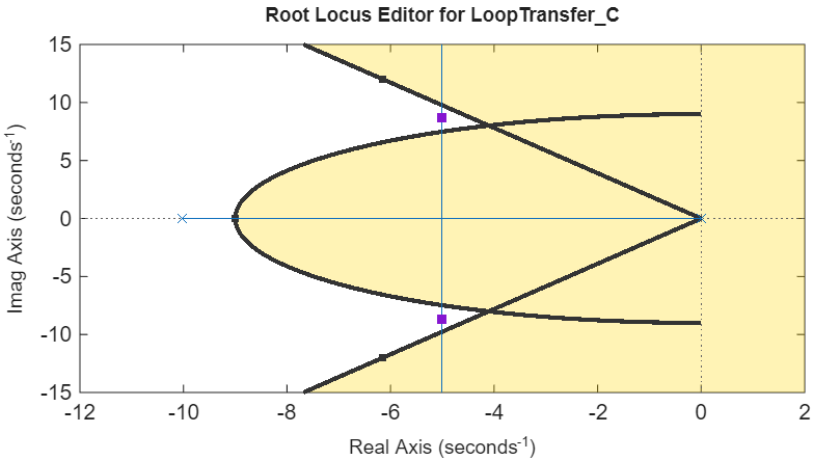
Determine the controller gain corresponding to a k_i that meets the specifications

$$t_r \leq 0.2 \text{ s},$$

less than 1% steady-state error, and overshoot less than 20%.

In theory, there should be a range of possible k_i gains that meet specifications (i)–(iii).

Root Locus Plot:



Where the yellow regions are where the time specifications are not satisfied. The closed loop poles are shown as $\circ$ for the k_i gain that met the specifications (found in problem 9). It lies within the region on the imaginary axis where the time specifications are satisfied.

6 Lab Discussion

14. Why do we need a model of the motor-speed control system, or plant?

We need a model of the plant in order to get an accurate transfer function for the system's closed and open loop responses. This allows us to design a controller that meets the desired time specifications. The transfer function can be used to predict the system's response to different inputs and to analyze the stability of the system.

15. What are the advantages of a P-controller? What are its disadvantages? Similarly, for an I-controller?

The P-controller was able to increase the speed of the system's response (decreases rise time) without any overshoot (except in cases of high k_p gain). However, it is not able to eliminate steady-state error without high gains which introduced oscillation in the system. The I-controller was able to eliminate steady-state error, but it introduced overshoot and oscillations in the system's response.

16. What are the advantages of a PI-controller? Justify your answer based on your experimental results.

The PI-controller was able to eliminate steady-state error while also reducing overshoot and oscillations in the system's response. It was able to balance the trade-off between rise time, overshoot, and steady-state error by adjusting the k_p and k_i gains. However, the effect of k_p gain on the input signal spread was still greater than desired as shown by the input signal spread of 3.09 V for the final k_p and k_i gains.

17. Why are the choices for the gains limited in practice?

They are limited by external disturbances and noise in the system, which can be amplified by high gains and result in instabilities in the system's response. Additionally, the physical limitations of the system, such as saturation of the actuator or sensor limits, can also limit the range of gains that can be used in practice.

7 Extra Credit

18. For your final k_p and k_i values in Question 10, rewrite the controller into the form

$$C(s) = \frac{K_c}{s}\left(s + \frac{1}{T_r}\right)$$

and plot the root locus showing the location of the closed-loop poles for your k_p and k_i gains, and corresponding K_c and T_r values.

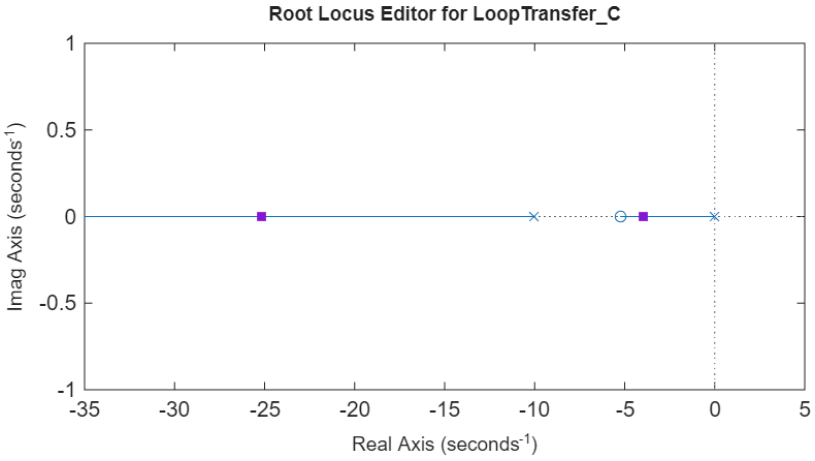
Compare the Control System Design response to your response. Does it meet the specifications?

Response:

$$C(s) = k_p + \frac{k_i}{s} = \frac{k_p s + k_i}{s} = \frac{k_p}{s}\left(s + \frac{k_i}{k_p}\right)$$

Thus, $K_c = k_p$ and $T_r = \frac{k_p}{k_i}$.

$$K_c = 0.08, \quad T_r = 0.19$$



The theoretical system response with the K_c and T_r values from the PI-controller design has the following time specifications:

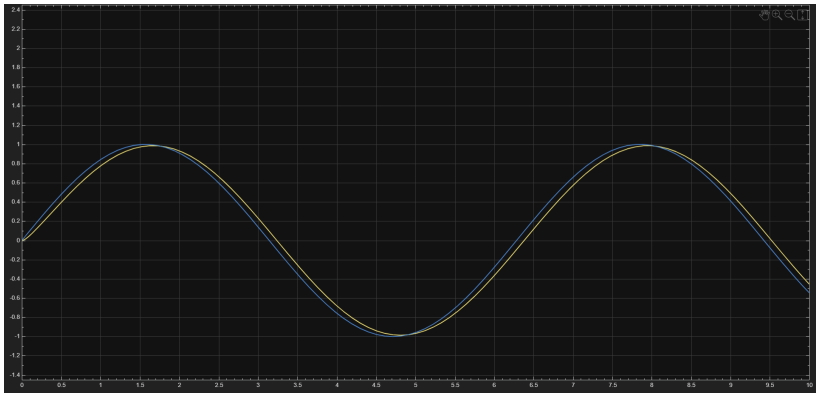
```
     RiseTime: 0.2597
TransientTime: 0.6666
 SettlingTime: 0.6666
  SettlingMin: 0.9009
  SettlingMax: 0.9995
    Overshoot: 0
   Undershoot: 0
         Peak: 0.9995
     PeakTime: 1.6079
```

Thus it has a rise time of 0.259 s, overshoot of 0.0%, and steady-state error of 0%. Generally, the response is similar to the experimental response.

19. Build a Simulink block diagram to track a sinusoidal signal with a frequency of

$$1 \text{ rad/s}.$$

How does your controller perform?



The controller tracks the sinusoidal signal well, with only a phase lag. Where the blue plot is the reference input and the yellow plot is the output.